

DRIVE SAFE

A dissertation submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF COMPUTER APPLICATIONS

of

University of Mysore



By

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DEPARTMENT OF BACHELOR OF COMPUTER APPLICATIONS
BGS FIRST GRADE COLLEGE
MYSURU-23
2023-2024

BGS FIRST GRADE COLLEGE.

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DEPARTMENT OF BACHELOR OF COMPUTER APPLICATIONS



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*This is to Certify that **Abishek P** bearing **U01CO21S0396** has completed his final semester project work entitled **Drive safe** as a partial fulfillment for the award of Bachelor of Computer Applications of University of Mysore during the year **2023-2024**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirement in respect of project work prescribed for the said degree.*

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We, student of final semester BCA of BGS First Grade College, Mysuru, hereby declare that the dissertation entitled “**Drive Safe**” has been independently carried at “**BGS First Grade College**”, Mysuru and submitted in partial fulfillment of the requirement for the award of **Bachelor of Computer Applications** affiliated to the **University of Mysore**, during the academic year 2023 – 2024. Further the matter embodied in the report is an original and bonafide work done by us.

To our knowledge this dissertation has not been submitted to any other college or university or published at any time prior to this.

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CHAPTER 1

INTRODUCTION

Chapter 1: Introduction

1.1 Topic Introduction:

Drowsiness, defined as the state of sleepiness when one needs to rest, can cause symptoms that have great impact over the performance of tasks: slowed response time, intermittent lack of awareness, or microsleeps (blinks with a duration of over 500 ms), to name a few examples. In fact, continuous fatigue can cause levels of performance impairment similar to those caused by alcohol. While driving, these symptoms are extremely dangerous since they significantly increase the probabilities of drivers missing road signs or exits, drifting into other lanes or even crashing their vehicle, causing an accident. For this work, our premise is the following: a camera mounted on a vehicle will record frontal images of the driver, which will be analyzed by using artificial intelligence (AI) techniques, such as deep learning, to detect whether the driver is drowsy or not. By using that information, the system will be able to alert the driver and prevent accidents. Given that the ADAS will have different functionalities integrated, one of the restrictions imposed to the module presented in this work will be to avoid the activation of false alarms that may distract the driver and cause him or her to turn off the ADAS. Thus, the main novelty of this work is the use of a non-intrusive system that is capable of detecting fatigue from sequences of images, which at the moment is an open problem. In most of the available works, the experimental methodology consists of extracting and classifying individual frames from each video and verifying whether the classification.

is a marked reduction in the new proposed model in comparison to the benchmark A sleepy driver is arguably much more dangerous on the road than the one who is speeding as he is a victim of microsleeps. Automotive researchers and manufacturers are trying to curb this problem with several technological solutions that will avert such a crisis. This article focuses on the detection of such micro sleep and drowsiness using neural network based methodologies. Our previous work in this field involved using machine learning with multi-layer perceptron to detect the same. In this paper, accuracy was increased by utilizing facial landmarks which are detected by the camera and that is passed to a Convolutional Neural Network (CNN) to classify drowsiness. The achievement with this work is the capability to provide a lightweight alternative to heavier classification models with more than 90% for the category without glasses, more than 80% for the category night without glasses. On average, more than 83% of accuracy was achieved in all categories. Moreover, as for model size, complexity and storage, there model where the maximum size is 75 KB.

1.2 Objectives:

- The main objective of this project to build model using deep learning techniques which helps to identify whether the person is drowsy or not

- The application helps in detecting eye of the person and alarm if the eye is closed for 10 seconds
- The objective of this project is to identify more accuracy result.

1.3 Motivation for the project:

As there is always increasing the death rate in the accident rate most of the accidents are happening due to the driver drowsing in highway. To overcome from this solution in the existing system we are going to build this model which helps in identifying the driver drowsy.

1.4 Project overview:

Driver drowsiness is among the leading causal factors in traffic accidents occurring worldwide. This paper describes a method to monitor driver safety by analyzing information related to fatigue using two distinct methods: eye movement monitoring and bio-signal processing. A monitoring system is designed where it receives sensory data via wireless sensor network and further processes the data to indicate the current driving aptitude of the driver. It is critical that several sensors are integrated and synchronized for a more realistic evaluation of the driver behaviour. The sensors applied include a video sensor to capture the driver image and a bio-signal sensor to gather the driver photoplethysmography signal. A warning alarm is sounded if driver fatigue is believed to reach a defined threshold. The manifold testing of the system demonstrates the practical use of multiple features, particularly with discrete methods, and their fusion enables a more authentic and ample fatigue detection.

1.5 LITERATURE SURVEY:

In various approaches steps are taken for facial landmark detection, object tracking and methods for driver drowsiness detection can be used in two ways either in contact approaches or in non-contact approaches. The nature of the techniques largely depends on the application domain.

Many researchers have attempted various technologies such as monitoring of underlying patterns in steering, monitoring vehicle position in lane, monitoring the eye/face of the driver, physiological measurement etc. It is seen that most approaches use driver eye/face monitoring, physiological measurement. The research done in 2019 uses driver eye/face monitoring which creates a "DriCare" alert for the driver. The system is efficient and its efficiency can be validated using a public driver drowsiness recognition dataset. Some of the research works which made the evolution to proposed work are depicted as follows.

A. Facial Landmarks Recognition Facial landmarks are an important, yet challenging phase in drowsiness detection. It has been applied to solve problems like alignment of face, estimation of head pose, swapping of face, blink detection etc. It usually used to pinpoint and characterize significant areas of the face, eyebrows, nose, and mouth. The research work by

the authors Y. Sun, X. Wang and X. Tang proposes a new methodology for approximation of the locations of facial key points with convolutional networks having three carefully designed levels. There are two benefits: first, the texture context data over the whole face is applied to find every key point. Second, for the reason that networks are skilled to predict all of the key points simultaneously, the geometric constraints amongst key points are implicitly encoded. The approach consequently can keep away from local minimum arising due to ambiguity and information corruption in difficult image samples because of occlusions, large pose variations and extreme lightings. Various network architectures are critical for precise and robust facial point detection. Results have been shown in three levels of convolutional networks: initial detection, tuned outcomes with the second and third levels of networks which improve the accuracy. Benefits are high performance convolutional networks. It also improves accuracy of modern-day methods and latest industrial software and additionally incredibly sturdy preliminary estimations. Drawback is that regionally sharing weights of neurons on the same map develops the performance but does not work well on images such as faces when globally sharing weights, and also is not precise under some exceptional poses and expressions.

B. Tracking of Visual Object Tracking of visual objects refers to tracing the trajectory of a single object across all frames of a video such that we are given its location only in the first frame. It is one of the major tasks in Computer Vision. Research work by D. S Bolme, J. R. Beveridge, B. A. Draper and Y. M. Lui, investigated a simpler tracking strategy. A novel correlation filter, a Minimum Output Sum of Squared Error (MOSSE) filter, which creates robust correlation filters when initialized using a single frame, is introduced. By using modified ASEF, UMACF or MOSSE filters good performance can be obtained under variations in rotation, scale, lighting and partial occlusion. The strength of a correlation peak can be used to detect occlusions or tracking failure, to avoid the online update and to reacquire the trace if the item reappears with a similar form by using "Peak-to-Side" lobe ratio. The subsequent algorithm can be just as precise and is much quicker. The advanced correlation filters are as good as complex trackers and the filter- based tactic is more than 20 times faster and can process 669 frames in a second. The end result is that they are not particularly sturdy to deviations in appearance of the target and fail on perplexing tracking problems. Research by M. Danelljan, G. Häger, F. S. Khan and M. Felsberg, explore the problem of exact and robust scale approximation for visual tracking in real-time. They have adopted the Discriminative Scale Space Tracker (DSST), which learns distinct correlation filters for unambiguous translation and scale assessment. Strategies are proposed to diminish the computational cost of tracking methods. Thus, we get a larger target search space without compromising real time performance. The tracking performance is significantly improved and speed increases two-fold.

C. Detection of Driver's Drowsiness There are two ways of detecting drowsiness; they are contact methods and non-contact methods. In contact approaches, the driver needs to wear or touch some physical parameter to measure level of drowsiness and on other hand in noncontact approaches, the driver does not need contact physical objects. Non-contact methods have lower cost and are easier. Research by S.J.Jung, H.-S.Shin and W.-Y. Chung, explores ways to monitor a driver's health state by using embedded electrocardiogram (ECG) sensors with electrically conductive fabric electrodes on the steering wheel. The ECG signals are sampled

at 100Hz from palms and transferred wirelessly to a remote station linked to a server computer in a private area network, to check driver's health condition, fatigue, and drowsiness state from physiological variations in biomedical signals. However, the pipeline was followed by a contact approach which may not be user-friendly and also time consuming. Research by G. Li, B.L.Lee and W.-Y. Chung, is based on physiological signals among them directly related to drowsiness is brain reflected by Electroencephalographic (EEG) signals. An SVM-based posterior probabilistic model and a smartwatch-based wearable EEG device were used. Drowsiness score is measured by probability value of $0 \sim 1$ rather than discrete class labels. But the EEG channel setting and the ground truth used are vague and the detection accuracy is not stated properly. Research by B. Warwick, N. Symons, X. Chen and K. Xiong, proposed a wireless wearable sensor used to build Driver Drowsiness systems. To design a drowsiness system there are two phases: collection of physiological data using the biosensor and analysis of measured data to identify the crucial factors related to the drowsiness.

In the second stage, there is designing of drowsiness detection algorithm and mobile app to alert drowsy people. It is found that a driver's heart rate and breathing rate are good indicators of drowsiness. Research by M. Omidyeganeh, A. Javadtalab and S. Shirmohammadi, captures facial appearance via a camera. The scheme contains four stages: Face extraction from image, eye detection, mouth detection, alert generation in drowsy state. Limitation of the system is that since non-contact approach is used here it depends on factors like light, camera and so on. The main aims of this paper are to overcome all drawbacks that are mentioned above.

CHAPTER 2

USER

APPLICATION

ENVIRONMENT

Chapter 2: User Application Environment

2.1 Python:

OVER VIEW OF PYTHON:

Python is an interpreted, high-level, general-purpose programming language. Created by Guido van Rossum and first released in 1991, Python's design philosophy emphasizes code readability with its notable use of significant whitespace. Its language constructs and object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects.



Python is dynamically typed and garbage-collected. It supports multiple programming paradigms, including procedural, object oriented, and functional programming. Python is often described as a "batteries included" language due to its comprehensive standard library.

Python was conceived in the late 1980s as a successor to the ABC language. Python 2.0, released in 2000, introduced features like list comprehensions and a garbage collection system capable of collecting reference cycles. Python 3.0, released in 2008, was a major revision of the language that is not completely backward-compatible, and much Python 2 code does not run unmodified on Python 3.

2.2 NumPy

NumPy is a library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays. The ancestor of NumPy, Numeric, was originally created by Jim

Hugunin with contributions from several other developers. In 2005, Travis Oliphant created NumPy by incorporating features of the competing Num array into Numeric, with extensive modifications. NumPy is open-source software and has many contributors.

The Python programming language was not originally designed for numerical computing, but attracted the attention of the scientific and engineering community early on. In 1995 the special interest group (SIG) matrix-sig was founded with the aim of defining an array computing package; among its members was Python designer and maintainer Guido van Rossum, who extended Python's syntax (in particular the indexing syntax) to make array computing easier.

2.3 PANDAS

Pandas is a software library written for the Python programming language for data manipulation and analysis. In particular, it offers data structures and operations for manipulating numerical tables and time series. It is free software released under the three-clause BSD license. The name is derived from the term "panel data", an econometrics term for data sets that include observations over multiple time periods for the same individuals. Its name is a play on the phrase "Python data analysis" itself. Wes McKinney started building what would become pandas at AQR Capital while he was a researcher there from 2007 to 2010.

2.4 OPEN CV

OpenCV (Open-source computer vision) is a library of programming functions mainly aimed at real-time computer vision. Originally developed by Intel, it was later supported by Willow Garage then Itseez (which was later acquired by Intel). The library is crossplatform and free for use under the open-source BSD license. OpenCV supports some models from deep learning frameworks like

TensorFlow, Torch, PyTorch (after converting to an ONNX model) and Caffe according to a defined list of supported layers. It promotes OpenVisionCapsules, which is a portable format, compatible with all other formats. Officially launched in 1999 the OpenCV project was initially an Intel Research initiative to advance CPU-intensive applications, part of a series of projects including real-time ray tracing and 3D display walls. The main contributors to the project included a number of optimization experts in Intel Russia, as well as Intel's Performance Library Team.

2.5 Dlib

Is a general-purpose cross-platform software library written in the programming language C++. Its design is heavily influenced by ideas from design by contract and component-based software engineering. Thus, it is, first and foremost, a set of independent software components. It is open-source software released under a Boost Software License.

Since development began in 2002, Dlib has grown to include a wide variety of tools. As of 2016, it contains software components for dealing with networking, threads, graphical user interfaces, data structures, linear algebra, machine learning, image processing, data mining, XML and text parsing, numerical optimization, Bayesian networks, and many other tasks. In recent years, much of the development has been focused on creating a broad set of statistical machine learning tools and in 2009 Dlib was published in the *Journal of Machine Learning Research*. Since then, it has been used in a wide range of domains

2.6 PyCharm

PyCharm's smart code editor provides first-class support for Python, JavaScript, CoffeeScript, TypeScript, CSS, popular template languages and more. Take advantage of language-aware code completion, error detection, and on-the-fly code fixes. Use smart search to jump to any class, file or symbol, or even any IDE action or tool window. It only takes one click to switch to the declaration, super method, test, usages, implementation, and more. Use smart

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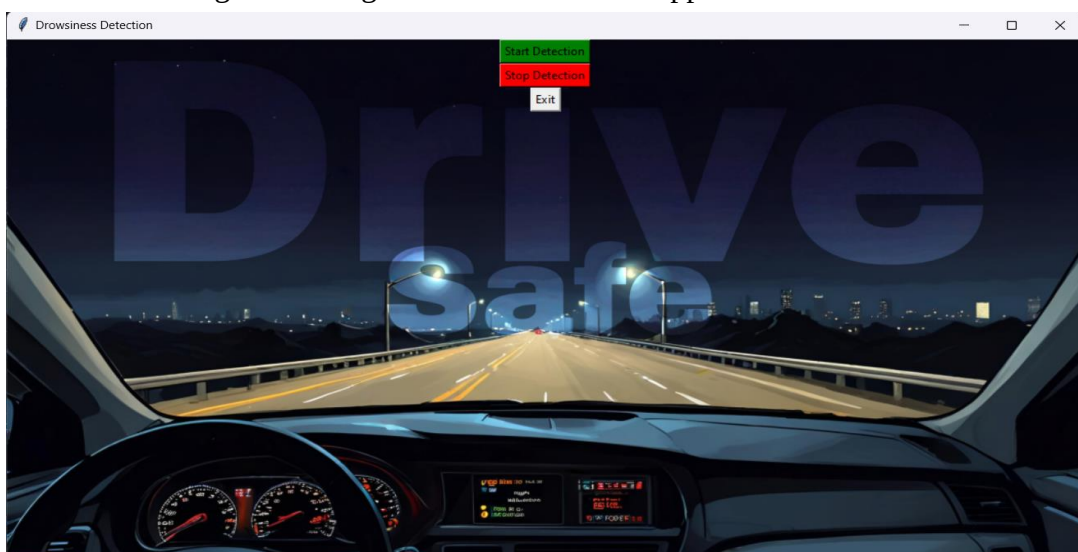
2.7 User Interface

The graphical user interface (GUI) is developed using the **tkinter** library. It provides the following functionalities:

- **Start Detection:** Begins the drowsiness detection process.
- **Stop Detection:** Halts the detection process.
- **Exit:** Closes the application.
- **About:** Displays information about the project, including its purpose and development details.

Key Features of the GUI

1. **Start Detection Button:**
 - This button initiates the drowsiness detection process, starting the video stream from the webcam and analyzing the video feed for signs of drowsiness.
2. **Stop Detection Button:**
 - This button stops the drowsiness detection process and closes the video stream.
3. **Exit Button:**
 - This button exits the application.
4. **About Button:**
 - This button opens a new window that provides information about the project, including its objectives and development details.
5. **Background Image:**
 - A background image enhances the visual appe



SCREENSHOT

Results

The system successfully detects drowsiness by analyzing eye and mouth movements in real-time. Alerts are triggered accurately based on predefined thresholds for EAR (Eye Aspect Ratio) and yawning frequency. The use of a laptop webcam ensures that the system is easy to set up and use.

The GUI allows users to interact with the system efficiently, providing clear and accessible controls to start and stop the detection process, as well as to exit the application and view information about the project. This user-friendly interface makes the drowsiness detection system practical for everyday use, potentially helping to prevent accidents caused by driver drowsiness.

CHAPTER 3

SYSTEM ANALYSIS

Chapter 3: System Analysis

3.1 Introduction to Design document

The Software Design will be used to aid in software development for android application by providing the details for how the application should built. Within the Software Design, specifications are narrative and graphical documentation of the software design for the project includes use case models, sequence diagrams and other supporting requirement information.

3.2 Scope

This software Design Document is for a base level system, which will work as a proof of concept for the use of building a system that provides a base level of functionality to show feasibility for large-scale production use. The software Design Document, the focus placed on generation of the documents and modification of the documents. The system will used in conjunction with other pre-existing systems and will consist largely of a document interaction faced that abstract document interactions and handling of the document objects. This Document provides the Design specifications of smart translation.

3.3 Hardware/Software Requirement

3.3.1 Hardware Requirements

- Processor : Intel i3 2.53GHz
- Hard Disk : 40GB
- RAM : 4GB or above

3.3.2 Software Requirements

- Operating System : windows 7 and above
- Language : Python
- IDE : PyCharm

3.4 FUNCTIONAL REQUIREMENTS:

Video Capture

The video capture unit used to record the video in real time of the frame containing the river face through a camera placed on the car dashboard.

Face Detection and Features Extraction

This unit receives the sampled video frame from the video capture unit. The images from the video capture unit are the RGB image and for the very dim light condition, we perform low-light image enhancement and noise elimination.

Fatigue Detection on Extracted Features

From the face detection unit, we get a sequence of eye and mouth image of the driver. Now from this extracted dataset, we can perform fatigue detection analysis on various facial features.

Alert Unit

The modelling of the alert unit is done on the format (r, t) , $\{r \geq 0, t \geq 0\}$ where r is the resultant value and t is the time. The whole model depends on the output value of the fatigue detection unit.

3.5 NON-FUNCTIONAL REQUIREMENTS:

Performance Requirements:

The product should uphold the end-client's necessities. The product is equipped for handling when the enormous quantities of records are given as information and furthermore, it should be intelligent and the deferrals included ought to be less. So, in each activity reaction of the framework, there are no prompt deferrals.

Safety and Security Requirements:

The system ought to be planned in a safe manner by applying wellbeing measures. Data transmission ought to be safely communicated to hubs with no progressions in data. A unique special case dealing with components ought to be set up to keep away from system mistakes.

3.6 FEASIBILITY STUDY

The feasibility study permits discovering answers for the issues of the undertaking. The arrangement is given what how resembles a fresh out of the new device resembles.

3.6.1 ECONOMICAL FESIBILITY

The general expense of the venture could be truly reasonable as we have utilized open-source programming like python. The investigation additionally assisted the organization with administering the sum that would be brought about before an undertaking is in the end support. This permitted organization to admirably achieve their cash to ensure the most productive venture is attempted. Economically the Scalability application is feasible as the developmental and maintenance charges are less than 15000 per year.

3.6.2 TECHNICAL FEASIBILITY

The software's utilized for implementing for the clients to move the insights in cross breed organizations. Without the race condition issues and invalid reservations, the data needs to move from source to objective. To keep away from the channel fluctuation and transfer speed confined to change measurements without issues to the objective.

3.6.3 OPERATIONAL FEASIBILITY

The task will be utilized by the association under various conditions. Anybody can work with this application as it upholds an easy-to-use approach. It gives graphical UIs to the client so the client can without much of a stretch connect with the framework. Clients no compelling reason to know about python and python IDE ideas to utilize the application. The application is planned so that it tends to be effectively carried out in any form of OS.

3.6.4 Social Feasibility

Social feasibility is a detailed study on how one interacts with others within a system or an organization. Social impact analysis is an exercise aimed at identifying and analyzing such impacts in order to understand the scale and reach of the project's social impacts. Schedule Feasibility is defined as the probability of a project to be completed within its scheduled time limits, by a planned due date. If a project has a high probability to be completed on-time, then its schedule feasibility is appraised as high.

3.6.5 SCHEDULE FEASIBILITY

The time schedule required for the development of this project is 5 months and it is very important since over-runs result in escalated projects costs and also hinders in the development of the other systems.

20 days – Required gathering

25 days – System designing

55 days – Implementation

35 days – Testing and deployment

CHAPTER 4

SYSTEM DESIGN

CHAPTER 4: SYSTEM DESIGN

The detailed configuration begins after the framework configuration stage is finished and the framework configuration has been guaranteed through the survey. The objective of this stage is to foster the inward rationale of every one of the modules distinguished during the framework plan.

In the framework plan, the emphasis is on distinguishing the modules, while during detailed design the attention is on planning the rationale for the modules. At the end of the day in framework, plan consideration is on which segments are required, while in detailed design how the segments can be executed in the product is the issue.

The planned movement is frequently partitioned into two separate stages framework plan and definite plan. Framework configuration is likewise called a high-level plan. At the primary level spotlight is on choosing which modules are required for the framework, the details of these modules, and how the modules ought to be interconnected. This is called a framework plan or high-level plan. In the subsequent level, the interior plan of the modules or how the particulars of the module can be fulfilled is chosen. This plan level is regularly called the detailed design or rationale plan.

4.1 HIGH LEVEL DESIGN

A data flow diagram is a graphical representation of the "flow" of data through an information system, modelling its process aspects. Often, they are a preliminary step used to create an overview of the system which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design). The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of the input data to the system, various processing carried out on these data, and the output data is generated by the system.

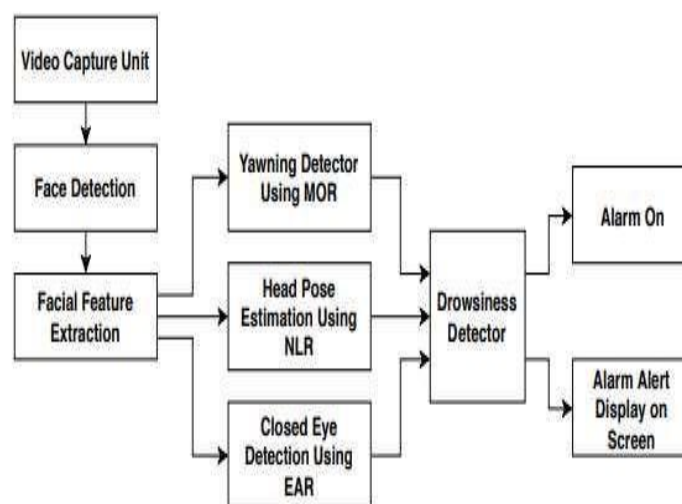


Fig: 4.1 High Level Design

A. Video Capture Unit

The video capture unit used to record the video in real time of the frame containing the river face through a camera placed on the car dashboard. The video is sampled with some frequency and the sampled frame is sent to the face detection unit.

B. Face Detection Unit and Features Extraction

This unit receives the sampled video frame from the video capture unit. The images from the video capture unit are the RGB image and for the very dim light condition, we perform low-light image enhancement and noise elimination. For improving the accuracy of our system, we eliminate the noise of the image before amplifying it through contrast enhancement. The above process is divided into two subtasks. At first, for denoising the image we apply the super pixel based adaptive noising and secondly, for amplifying the image luminance adaptive contrast enhancement is used. We need to denoise the image before contrast enhancement, so that the noise has been eliminated before its amplification through contrast enhancement. The above method increases the accuracy of our system significantly as it eliminates heavy noise, texture blurring and over-enhancement from the image which is then processed as accordingly. The image is changed to the grayscale image because for face detection we do not need the colour data. For face detection in the frame, we use the rapid object detection which uses the boosted need only the eyes and mouth part of the face for the feature set. For this first, we scaled the rectangle boxed in face image to 100*100 pixels and then we use the 80x30 rectangular pixels window for eye and 40x40 rectangular pixels window for the mouth. The x- and y coordinates of the rectangular window is (10, 20) and (30, 60) for the eye and mouth respectively. The above cascade of the classifier by Viola-Jones that works with the Haar-like features. The face detection method returns the abscissa, ordinates, length, and breadth of the rectangle boxed in the facial image. As we part is then transferred to the fatigue detection unit for further processing.

C. Fatigue Detection on Extracted Features

From the face detection unit, we get a sequence of eye and mouth image of the driver. Now from this extracted dataset, we can perform fatigue detection analysis on various facial features. These facial features include eyes (fast blinking or heavy eyes). The combined result of fatigue detection on these facial features is used to give the final result as to whether the driver is in fatigue or alert state. The images from face detection unit, in pixels, can contain a lot of features. If we perform as said in the previous unit we would get a feature vector of size 4000. This size can be further reduced with the use of Principle Component Analysis (PCA). PCA can be used to avoid the problem caused by highdimensionality as it compresses the data while minimizing the amount of information lost. It searches for a pattern in the data and reduces as much possibly correlated high dimensional variables. The compressed data set can now be divided into training set and test set. To classify if the driver is fatigued or not fatigued, we use Support Vector Classifier. This can also be correlated from the fact that Support Vector Classifier is highly efficient in working with the high dimensional feature vector. It is also very flexible in dealing with linearly as well as nonlinearly separable data sets. SVM is a supervised learning method used for classification and regression. SVM is also referred to as Maximum

Margin Classifier because it can maximize the geometric margin and minimize the empirical classification error simultaneously. During classification SVM creates a maximal separating hyperplane. Now, two parallel hyperplanes are constructed on each side of hyperplane that separates the data. Here an assumption is made that larger the distance between the two parallel planes better is the generalization error of classifier. SVM was used to implement this problem because of, binary nature of classification problem and the efficiency of SVM in working with high dimensional data and its flexibility in working with both linearly and nonlinearly separable data sets. Unlike other classification algorithms, SVM can be used in both linear and non-linear ways with the use of a kernel. In cases when we have a limited set of points in 228 many dimensions SVM tends to be very efficient because it can find linear separation in the data. SVM is also eliminates the drawbacks of outliers as it uses only relevant points to find a linear separation, also called support vectors. Now, we can train the classifier on the training set and can check its accuracy using test set. The test set and training set are completely different i.e., no two instances in both data sets are same. We do this so that our SVC model is always tested on a data which it has not seen before. This strategy provides a better picture of the generalized functioning of SVC. After the SVC model is trained, we can calculate the reduction accuracy of our model using cross validation. In cross-validation, we perform a number of iterations, wherein each iteration we give a different data subset to test set from the previously allotted set. Finally, the classifier would return 1 if the driver is found to be in fatigue state or return -1 if the driver is found to be in an alert state to the alert unit.

D. Alert Unit

The modelling of the alert unit is done on the format (r, t) , $\{r \geq 0, t \geq 0\}$ where r is the resultant value and t is the time. The whole model depends on the output value of the fatigue detection unit. The Support Vector Machine (SVM) classifies the image, whether they are fatigued or not fatigued. The representation of the classification output is either a +1 or -1 and this number is used by the alert unit in the following way. Specifically, the classification output is the input of a running sum that adds the consecutive output values and has a minimum value up to zero. This unit employs two threshold level, first one is to detect whether there is no or low fatigue level. The second threshold value is to mark the difference between low and high fatigue level. Whenever the resultant value goes above a specified threshold, it implies that fatigue is detected and the alert system goes active and takes action according to the detected fatigue level. For low fatigue level, the alarm rings for 10sec and after 10sec the system again checks the resultant value and performs the alert process accordingly. For high fatigue level, we can use more effective accident preventive measures like an automatic reduce the speed and ultimately stop the vehicle and, or water spray. This system can be used to track the fatigue level of a driver and detect the sleep onset with a safe margin.

4.2 DETAILED DESIGN

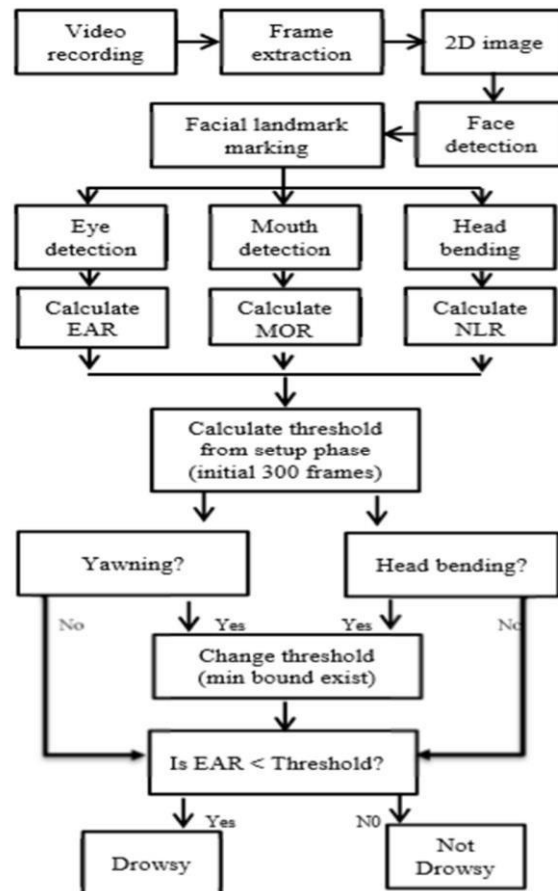


Fig: 4.2 Block diagram of drowsiness system

A block diagram of the driver drowsiness monitoring system has been depicted in Fig 4.2. At first, the video is recorded using a webcam. The camera will be positioned in front of the driver to capture the front face image. From the video, the frames are extracted to obtain 2-D images. Face is detected in the frames using histogram of oriented gradients (HOG) and linear support vector machine (SVM) for object detection. After detecting the face, facial landmarks like positions of eye, nose, and mouth are marked on the images. From the facial landmarks, eye aspect ratio, mouth opening ratio and position of the head are quantified and using these features and machine learning approach, a decision is obtained about the drowsiness of the driver. If drowsiness is detected, an alarm will be sent to the driver to alert him/her.

4.2.1 FACIAL LANDMARK MARKING

After detecting the face, the next task is to find the locations of different facial features like the corners of the eyes and mouth, the tip of the nose and so on. Prior to that, the face images should be normalized in order to reduce the effect of distance from the camera, non-uniform illumination and varying image resolution. Therefore, the face image is resized to a width of 500 pixels and converted to grayscale image. After image normalization, ensemble of regression trees is used to estimate the landmark positions on face from a sparse subset of pixel

intensities. In this method, the sum of square error loss is optimized using gradient boosting learning. Different priors are used to find different structures. Using this method, the boundary points of eyes, mouth and the central line of the nose are marked and the number of points for eye, mouth and nose are given in Table I. The facial landmarks are shown in Fig 4.3. The red points are the detected landmarks for further processing.

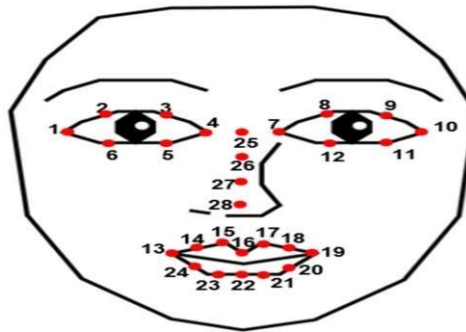


Fig: 4.3 Facial Land Marking

Table:

Parts	Landmark Points
Mouth	13-24
Right eye	1-6
Left eye	7-12
Nose	25-28

Table 4.1: Facial Land Marking Points

4.3 ALGORITHM

It consists of Video Capture Unit, Face Detection Unit, Head pose estimation system, Facial feature Extraction, closed eye detection, yawning detection, drowsiness detection, Alert Unit.

The general flow of drowsiness detection algorithm is given below.

- First, we will setup a camera that monitors a stream for face.
- If a face is found, we apply facial landmark detection and extract the eye regions
- If the eye aspect ratio indicates that the eyes have been closed for a sufficiently long enough amount of time, then sound an alarm to wake up the driver.
- If the mouth aspect ratio indicates that the yawning for a sufficiently long enough amount of time, then sound an alarm to wake up the driver.
- Estimate of head pose also detect the drowsiness of driver.

4.3.1 EYE ASPECT RATIO (EAR)

Mouth opening ratio is a parameter to detect yawning during drowsiness. Similar to EAR, it is calculated as

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

where p_i represents point marked as i in facial landmark and is the distance between points marked as i and j . Therefore, when the eyes are fully open, EAR is high value and as the eyes are closed, EAR value goes towards zero. Thus, monotonically decreasing EAR values indicate gradually closing eyes and it's almost zero for completely closed eyes (eye blink). Consequently, EAR values indicate the drowsiness of the driver as eye blinks occur due to drowsiness.

4.3.2 MOUTH OPENING RATIO (MOR):

Mouth opening ratio is a parameter to detect yawning during drowsiness. Similar to EAR, it is calculated as

$$MOR = \frac{(p_{15} - p_{23}) + (p_{16} - p_{22}) + (p_{17} - p_{21})}{3(p_{19} - p_{13})}$$

As defined, it increases rapidly when mouth opens due to yawning and remains at that high value for a while due to yawn (indicating that the mouth is open) and again decreases rapidly towards zero. As yawn is one of the characteristics of drowsiness, MOR gives a measure regarding driver drowsiness.

4.3.3 HEAD BENDING:

Due to drowsiness, usually driver's head tilts (forward or backward) with respect to vertical axis. So, from the head bending angle, driver drowsiness can be detected. As the projected length of nose on the camera focal plane is proportional to this bending, it can be used as a measure of head bending. In normal condition, our nose makes an acute angle with respect to focal plane of the camera. This angle increases as the head moves vertically up and decreases on moving down. Therefore, the ratio of nose length to an average nose length while awake is a measure of head bending and if the value is greater or less than a particular range, it indicates head bending as well as drowsiness. From the facial landmarks, the nose length is calculated and it is defined as

$$NLR = \frac{\text{nose length}(p_{28} - p_{25})}{\text{average nose length}}$$

The average nose length is computed during the setup phase of the experiment as described in the next sub-section.

4.4 CODE:

```

import tkinter as tk
from PIL import Image, ImageTk
from threading import Thread
import cv2
import numpy as np
import imutils
from scipy.spatial import distance as dist
from imutils.video import VideoStream
from imutils import face_utils
import dlib
import os
import time
from pygame import mixer

mixer.init()
mixer.music.load(r"C:\\Users\\Lenovo\\Downloads\\Project-Alarm-
[AudioTrimmer.com].mp3")

def alarm(msg):
    global alarm_status
    global alarm_status2
    global saying

    while alarm_status:
        print('call')
        s = 'speak "' + msg + '"'
        os.system(s)

    if alarm_status2:
        print('call')
        saying = True
        s = 'speak "' + msg + '"'
        os.system(s)
        saying = False

def eye_aspect_ratio(eye):
    A = dist.euclidean(eye[1], eye[5])
    B = dist.euclidean(eye[2], eye[4])

    C = dist.euclidean(eye[0], eye[3])

    ear = (A + B) / (2.0 * C)

```

```

    return ear

def final_ear(shape):
    (lStart, lEnd) = face_utils.FACIAL_LANDMARKS_IDXS["left_eye"]
    (rStart, rEnd) = face_utils.FACIAL_LANDMARKS_IDXS["right_eye"]

    leftEye = shape[lStart:lEnd]
    rightEye = shape[rStart:rEnd]

    leftEAR = eye_aspect_ratio(leftEye)
    rightEAR = eye_aspect_ratio(rightEye)

    ear = (leftEAR + rightEAR) / 2.0
    return (ear, leftEye, rightEye)

def lip_distance(shape):
    top_lip = shape[50:53]
    top_lip = np.concatenate((top_lip, shape[61:64]))

    low_lip = shape[56:59]
    low_lip = np.concatenate((low_lip, shape[65:68]))

    top_mean = np.mean(top_lip, axis=0)
    low_mean = np.mean(low_lip, axis=0)

    distance = abs(top_mean[1] - low_mean[1])
    return distance

EYE_AR_THRESH = 0.3
EYE_AR_CONSEC_FRAMES = 30
YAWN_THRESH = 20
alarm_status = False
alarm_status2 = False
saying = False
COUNTER = 0
vs = None

detector =
cv2.CascadeClassifier("C:\\Users\\Lenovo\\PycharmProjects\\Final\\.venv\\Lib\\site-
packages\\cv2\\data\\haarcascade_frontalface_default.xml") # Faster but less accurate
predictor = dlib.shape_predictor('C:\\Users\\Lenovo\\PycharmProjects\\Final\\.venv\\Lib\\site-
packages\\cv2\\data\\shape_predictor_68_face_landmarks.dat')

def start_detection():
    global detection_thread
    global vs

```

```

vs = VideoStream(src=0).start()
time.sleep(1.0)
detection_thread = Thread(target=detect_drowsiness)
detection_thread.daemon = True
detection_thread.start()

def stop_detection():
    global vs

    if vs is not None:
        vs.stop()
        cv2.destroyAllWindows()

def detect_drowsiness():
    global alarm_status
    global alarm_status2
    global saying
    global COUNTER
    global vs

    while True:
        frame = vs.read()
        frame = imutils.resize(frame, width=450)
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

        rects = detector.detectMultiScale(gray, scaleFactor=1.1,
                                           minNeighbors=5, minSize=(30, 30),
                                           flags=cv2.CASCADE_SCALE_IMAGE)

        for (x, y, w, h) in rects:
            rect = dlib.rectangle(int(x), int(y), int(x + w), int(y + h))

            shape = predictor(gray, rect)
            shape = face_utils.shape_to_np(shape)

            eye = final_eye(shape)
            ear = eye[0]
            leftEye = eye[1]
            rightEye = eye[2]

            distance = lip_distance(shape)

            leftEyeHull = cv2.convexHull(leftEye)
            rightEyeHull = cv2.convexHull(rightEye)
            cv2.drawContours(frame, [leftEyeHull], -1, (0, 255, 0), 1)
            cv2.drawContours(frame, [rightEyeHull], -1, (0, 255, 0), 1)

            lip = shape[48:60]

```

```

cv2.drawContours(frame, [lip], -1, (0, 255, 0), 1)

if ear < EYE_AR_THRESH:
    COUNTER += 1

    if COUNTER >= EYE_AR_CONSEC_FRAMES:
        if not alarm_status:
            alarm_status = True
            t = Thread(target=alarm, args=('wake up sir',))
            t.daemon = True
            t.start()

        cv2.putText(frame, "DROWSINESS ALERT!", (10, 30),
                     cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)
        mixer.music.play()

    else:
        COUNTER = 0
        alarm_status = False

if (distance > YAWN_THRESH):
    cv2.putText(frame, "Sleepy!.....", (10, 30),
                cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)
    mixer.music.play()
    if not alarm_status2 and not saying:
        alarm_status2 = True
        t = Thread(target=alarm, args=('take some fresh air sir',))
        t.daemon = True
        t.start()
    else:
        alarm_status2 = False

cv2.putText(frame, "EAR: {:.2f}".format(ear), (300, 30),
            cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)
cv2.putText(frame, "YAWN: {:.2f}".format(distance), (300, 60),
            cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)

cv2.imshow("Frame", frame)
key = cv2.waitKey(1) & 0xFF

if key == ord("q"):
    break

def exit_application():
    root.destroy()

root = tk.Tk()
root.title("Drowsiness Detection")

```

```

# Set the size of the window
root.geometry("1050x600")

# Load background image
background_image = Image.open("C:\\Users\\Lenovo\\Desktop\\Picsart_24-05-04_13-42-59-130.png")
background_photo = ImageTk.PhotoImage(background_image)

# Create a label to display the background image
background_label = tk.Label(root, image=background_photo)
background_label.place(x=0, y=0, relwidth=1, relheight=1)

start_button = tk.Button(root, text="Start Detection", command=start_detection, bg="green")
start_button.pack()

stop_button = tk.Button(root, text="Stop Detection", command=stop_detection, bg="red")
stop_button.pack()

exit_button = tk.Button(root, text="Exit", command=exit_application)
exit_button.pack()

def show_about_info():
    about_window = tk.Toplevel(root)
    about_window.title("About Drowsiness Detection Project")
    about_window.geometry("400x200")

    about_frame = tk.Frame(about_window)
    about_frame.pack(fill="both", expand=True)

    about_label = tk.Label(about_frame, text="Drive Safe Project\n\n"
                                     "This project detects drowsiness \n in real-time using computer \n
vision techniques. "
                                     "It monitors the eyes and mouth \n of the person to detect signs \n
of drowsiness. "
                                     "When drowsiness is detected, it \n alerts the user with sound and
\n visual representations.\n\n"
                                     "Developed by [BGS Students 23/24]")
    about_label.pack(fill="both", expand=True)

# Add About button with custom appearance
about_button = tk.Button(root, text="About", command=show_about_info, bg="#4CAF50",
fg="white",
                        font=("Helvetica Bold", 12), relief="raised", borderwidth=3)
about_button.pack(side="bottom", anchor="sw", padx=10, pady=10)

root.mainloop()

```

CHAPTER 5

SYSTEM

IMPLEMENTATION

CHAPTER 5: SYSTEM IMPLITATION

5.1 INTRODUCTION

Implementation is the way toward changing over another or an updated framework plan into an operational one. The goal is to put the new or overhauled framework that has been tried into activity while holding expenses, dangers, and individuals bothering to the base. A basic part of the execution cycle is to guarantee that there will be no disturbing the working of the association. The best strategy for acquiring control while embedding any new framework is to utilize all-around arranged tests for testing every single new program. Before creation documents are utilized to test live information, text records should be made on the old framework, replicated over to the new framework, and utilized for the underlying trial of each program.

Another factor to be considered in the execution stage is the obtaining of the equipment and programming. When the product is created for the framework and testing is completed, it is then the way toward making the recently planned framework completely operational and reliable in execution.

Implementation is the most significant stage in accomplishing a fruitful framework and giving the client certainty that the new framework is useful and viable. Execution of an altered application to supplant a current one. This sort of discussion is moderately simple to deal with, give there are no significant changes in the framework.

5.2 PARALLEL CONVERSION TYPE OF IMPLEMENTATION

In this type of implementation both the current system and the proposed system run in parallel. This happens till the user gets the complete confidence on the proposed system and hence cuts of the current system.

5.2.1 PHASE - IN METHOD OF IMPLEMENTATION

In this sort of implementation, the proposed framework is presented stage-by stage. This lessens the danger of vulnerability of the proposed framework.

Each program is tried independently at the hour of improvement utilizing the information and has checked that this program connected together in the manner indicated in the determination of the program, the PC framework, and its current circumstance is tried as per the general inclination of the client. The framework that has been created is acknowledged and ends up being acceptable for the client. Thus, the framework will be executed very soon. A straightforward working system is incorporated so the client can comprehend the various capacities unmistakably and rapidly.

At first, as an initial step, the executable type of the application is to be made and stacked in the normal worker machine which is open to the whole client and the worker is to

be associated with an organization. The last stage is to archive the whole framework which gives parts and the working methods of the framework.

Implementation is the phase of the venture when the hypothetical plan is transformed out into a functioning framework. Subsequently it very well may be viewed as the most basic stage in accomplishing a fruitful new framework and in giving the client, the certainty that the new framework will work and be successful.

The implementation stage includes cautious arranging, examination of the current framework and its imperatives on execution, planning of techniques to accomplish changeover, and assessment of changeover strategies.

Implementation is the way toward changing over another framework plan into activity. The stage center on client preparation, site arrangement, and record transformation for introducing an up-and-comer framework. The significant factor that ought to be considered here is that the change ought not to disturb the working of the association.

5.3 IMPLEMENTATION METHODOLOGY OF THE PROJECT

The project is implemented in modular approach. Each module is coded as per the requirements and tested and this process is iterated till the all the modules have been thoroughly implemented.

5.4 Methodology:

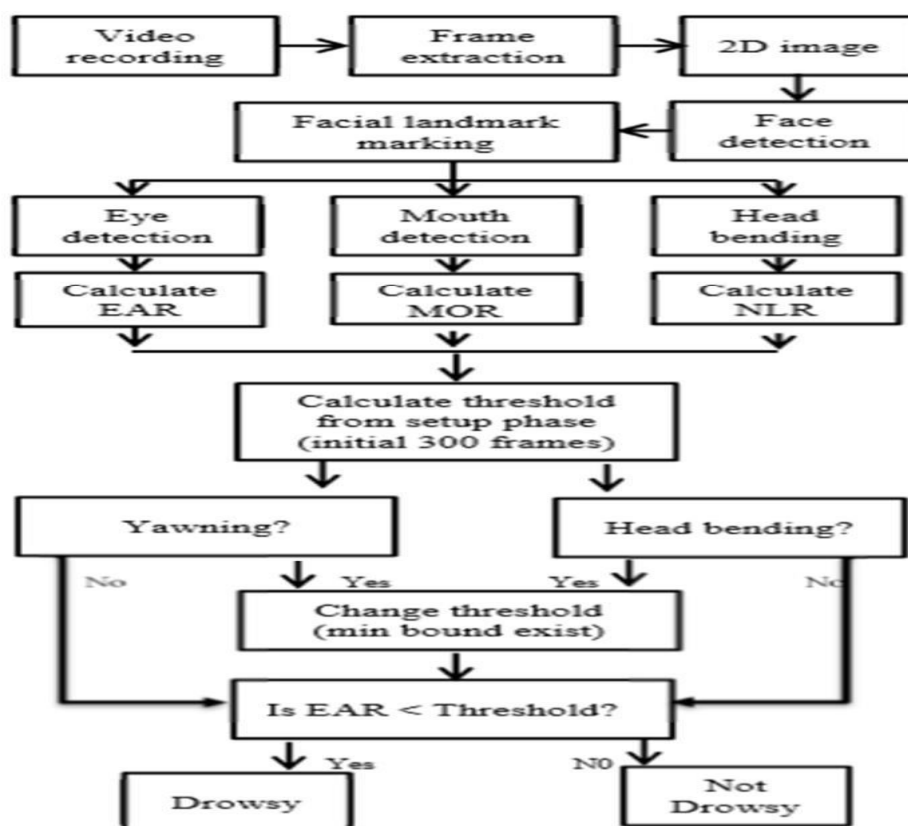


Fig 5.4.1: Block Diagram of Proposed System

A block diagram of the driver drowsiness monitoring system has been depicted in Fig 6.4.1 At first, the video is recorded using a webcam. The camera will be positioned in front of the driver to capture the front face image. From the video, the frames are extracted to obtain 2-D images. Face is detected in the frames using histogram of oriented gradients (HOG) and linear support vector machine (SVM) for object detection. After detecting the face, facial landmarks like positions of eye, nose, and mouth are marked on the images. From the facial landmarks, eye aspect ratio, mouth opening ratio and position of the head are quantified and using these features and machine learning approach, a decision is obtained about the drowsiness of the driver. If drowsiness is detected, an alarm will be sent to the driver to alert him/her.

CHAPTER 6

TESTING

CHAPTER 6: TESTING

Testing is the process of debugging the programs. The purpose is to find out the errors in the product development time. The objective of the product testing is to convince framework designer and clients that the product is adequate for operational utilize. Testing is a procedure proposed to have trust in the product. Testing is a preparation of activities that can be in order ahead of time and directed analytically and is frequently referred to as confirmation and approval.

6.1 IMPORTANCE OF TESTING

- Testing is a procedure of executing a program with the plan of looking through an error.
- Testing is a process is to debug the programs and produce the mistakes in the programs.
- The employment of testing is to ensure that the product does its activities accurately and can be utilized as a part of creation.
- Gaining certainty about the level of quality.
- Providing data for decision making.
- Ensure that all the functionalities are completed successfully.

6.2 BLACK BOX TESTING

The STD system is tested to all below test objectives

Test Objectives

- All features on face must detect properly, which leads to the corresponding output.
- Application must be activated as soon as webcam is On.
- After opening of webcam Application must not be late to redirect.

Feature Testing

- Confirm that the application made by client is the valid application.
- All connections would take the client to redirect page as supposed to take to the wrong page.
- No indistinguishable video/images must be permitted in the framework.

Regression Testing

If the driver is not visible to the webcam the application not able to detect drowsiness.

6.3 WHITE BOX TESTING

In the unit testing the following modules are tested.

- The button directs to proper and correct page.

- The dataflow of files between application and webcam.
- Testing the pre uploaded results to train the system.
- Testing the webcam working.
- Testing application working

Integration testing

- Checked whether the pre tested results are uploaded to the system.
- Checked whether the data flow is proper between system and webcam.
- Single click on the button redirects to the webcam.

6.4 TESTING UNITS

Testing is to be performed in this project on the following modules.

- Video capture.
- Face detection.
- Eye detection.
- Mouth detection
- Head detection

Video Detection:

Test case no	Test cases	Condition	Expected output	Actual output
01	Install the web cam in car	Web cam should work properly	Accept	Webcam installed
02	Video Detection	Driving video should detect	Accept	Video Detected Successfully
03	Install drowsiness detection application	Valid application should be installed	Valid installation	Installation of application is successful

Table 6.1 : Video Detection

Face Detection:

Test case no	Test cases	Condition	Expected output	Actual output
01	Driver face detection	Proper position of webcam	Driver face detection	Driver face detected
02	Detection of facial land mark	Proper land marks detection	Location of facial features detection	Land marks of face detected
03	Detection of positive features	Specific feature detection	Positive features detection	Positive features detected

Table 6.2: Face detection

Eye Detection:

Test case no	Test cases	Condition	Expected output	Actual output
01	Eye detection	Detecting eye feature	Eye landmark detection	Eye landmark detected
02	Detecting eye corner points	Land mark of corner point	Corner points detection	Land mark of eye corner point detected
03	Detecting closing and opening of eye	Land mark of eye lashes	Closing and opening of eye land mark detection	Land marks of closing and opening of eye detected

Table 6.3: Eye detection

Mouth Detection:

Test case no	Test cases	Condition	Expected output	Actual output
01	Mouth detection	Detecting mouth feature	Mouth landmark detection	Mouth landmark detected
02	Detecting mouth corner points	Land mark of corner point	Corner Points detection	Land mark of mouth corner point detected
03	Detecting and closing of opening mouth	Land mark of Mouth while yawning	Closing and opening of mouth land mark detection	Land marks of closing and opening of mouth detected

Table 6.4: Mouth detection

Head Bending:

Test case no	Test cases	Condition	Expected output	Actual output
01	Tip of nose detection	Landmark of nose tip	Land mark of tip of the nose detection	Land mark detected
02	Downward and upward of nose detection	Landmark of nose tip when it is downward and upward	Bending of Head detection	Landmarks detected
03	Head bending detection	Land marks of nose when it downwards	Landmarks of head bending detection	Head bending detected

Table 6.4: Head bending

CHAPTER 7

CONCLUSION

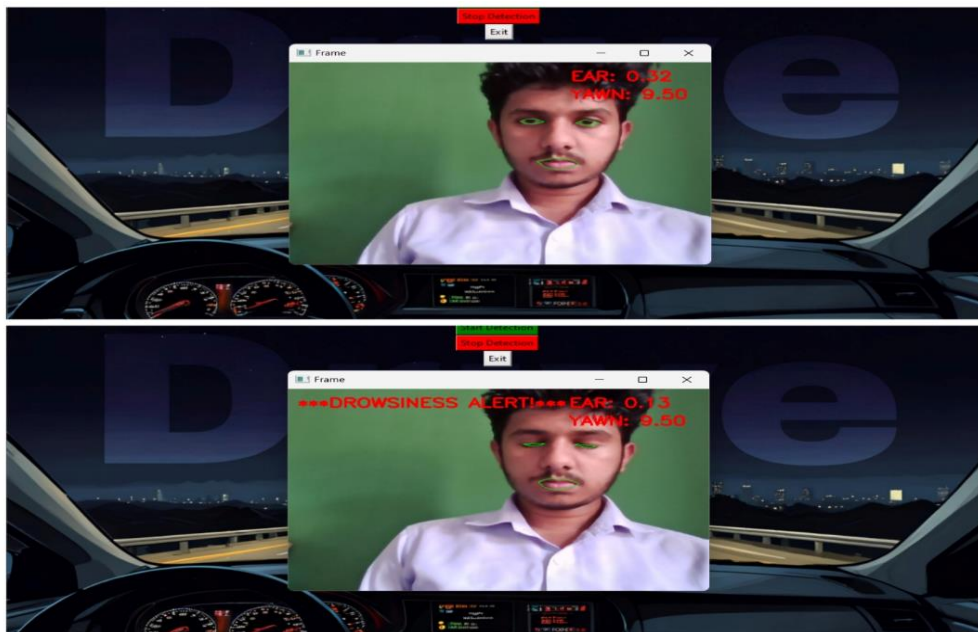
CHAPTER:7 CONCLUSION

7.1 Conclusion

This system works only on the required part of face image i.e., eyes and mouth, rejects the rest. This step decreases the unnecessary features in the feature set. Eye and mouth detection is less accurate than the face detection, that's why we use the face detection to get the eye and mouth image part of the driver. Estimating the head position of the driver will detect the drowsiness of driver. It makes the system optimized in the context of time and accuracy. The division of system alert unit into three units is an efficient way to alert the driver. It works in such a way that the driver is not subjected to sudden attack, which may lead to an accident.

7.2 Future Enhancements

As future enhancement Pedestrian detection is an essential and significant task in any intelligent video surveillance system, as it provides the fundamental information for semantic understanding of the video footages. It has an obvious extension to automotive applications due to the potential for improving safety systems.



SCREENSHOT

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